

Title

Demonstration of Import, Export, Dump, and Restore Operations in MongoDB

Business Domain

Freelance Marketplace Management System

AIM

To demonstrate MongoDB import, export, backup, restore, deletion, and indexing operations using a Freelance Marketplace Management System.

BUSINESS DOMAIN

The selected business domain is a Freelance Marketplace Management System.

The system manages freelance projects posted by clients and completed or managed by freelancers. The project information includes project ID, project title, category, client name, freelancer name, budget, status, and rating.

1. Import the prepared dataset into the appropriate MongoDB collection using the mongoimport utility.
2. Append Additional Records: Create a separate JSON or CSV file containing at least five additional records relevant to the selected business domain. Import the file into the existing collection to append the new records.

```
(base) sshailmaadan@OshailsMacBook - % nano ~/Downloads/projects.csv
(base) sshailmaadan@OshailsMacBook - % cat ~/Downloads/projects.csv
project_id,project_title,client_name,category,budget,status,freelancer_name
P101,Website Development,TechNova,Web Development,45000,Completed,Aarav Sharma
P102,Logo Design,GreenLeaf,Graphic Design,8000,In Progress,Meha Kapoor
P103,Mobile App UI,FitLife,UI UX Design,25000,Completed,Rohan Mehta
P104,Content Writing,EduWorld,Content Writing,12000,Open,Priya Singh
P105,Python Automation,DataCorp,Programming,30000,In Progress,Karan Malhotra

(base) sshailmaadan@OshailsMacBook - % mongoimport --db freelanceDB --collection projects --type csv --headerline --file ~/Downloads/projects.csv
2026-08-15T17:58:44.300+0530   connected to: mongodb://localhost/
2026-08-15T17:58:44.330+0530   5 document(s) imported successfully. 0 document(s) failed to import.
(base) sshailmaadan@OshailsMacBook - % mongosh
Current Mongosh Log ID: 6a805ba3b7b4aa421f8be58a
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.9.2
Using MongoDB:     6.3.7
Using Mongosh:      2.9.2
mongosh 2.10.0 is available, run brew upgrade mongosh to update

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
[ The server generated these startup warnings when booting
2026-08-15T13:44:00.902+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> use freelanceDB
switched to db freelanceDB
freelanceDB> db.projects.countDocuments()
5
```

3. Verify the data in the collection by performing the following:

- a. Display the total number of documents in the collection.

```
freelanceDB8> db.projects.countDocuments()
```

5

- b. Display a sample document from the collection.

```
freelanceDB8> db.projects.findOne()
{
  _id: ObjectId('6a885b7cd84b74cccf6831dd'),
  project_id: 'P184',
  project_title: 'Content Writing',
  client_name: 'EduWorld',
  category: 'Content Writing',
  budget: 12000,
  status: 'Open',
  freelancer_name: 'Priya Singh'
}
freelanceDB8> db.projects.find()
[
  {
    _id: ObjectId('6a885b7cd84b74cccf6831dd'),
    project_id: 'P184',
    project_title: 'Content Writing',
    client_name: 'EduWorld',
    category: 'Content Writing',
    budget: 12000,
    status: 'Open',
    freelancer_name: 'Priya Singh'
  },
  {
    _id: ObjectId('6a885b7cd84b74cccf6831de'),
    project_id: 'P183',
    project_title: 'Mobile App UI',
    client_name: 'FitLife',
    category: 'UI UX Design',
    budget: 25000,
    status: 'Completed',
    freelancer_name: 'Rohan Mehta'
  },
  {
    _id: ObjectId('6a885b7cd84b74cccf6831df'),
    project_id: 'P181',
    project_title: 'Website Development',
    client_name: 'TechNova',
    category: 'Web Development',
    budget: 45000,
    status: 'Completed',
    freelancer_name: 'Aarav Sharma'
  },
  {
    _id: ObjectId('6a885b7cd84b74cccf6831e0'),
    project_id: 'P182',
    project_title: 'Logo Design',
    client_name: 'GreenLeaf',
    category: 'Graphic Design',
    budget: 8000,
    status: 'In Progress',
    freelancer_name: 'Neha Kapoor'
  },
  {
    _id: ObjectId('6a885b7cd84b74cccf6831e1'),
    project_id: 'P185',
    project_title: 'Python Automation',
    client_name: 'DataCorp',
    category: 'Programming',
    budget: 30000,
    status: 'In Progress',
    freelancer_name: 'Karan Malhotra'
  }
]
```

- c. Display the newly added records.

```
freelanceDB> db.projects.find({
[
  {
    project_id: {
      $gte: 106,
      $lte: 110
    }
  }
}).pretty()
[
  {
    _id: ObjectId('6a8861feab527df55efa3636'),
    project_id: 107,
    project_title: 'WordPress Business Website',
    category: 'Web Development',
    client_name: 'Pooja Arora',
    freelancer_name: 'Vivek Sharma',
    budget: 22000,
    status: 'Completed',
    rating: 4.5
  },
  {
    _id: ObjectId('6a8861feab527df55efa3637'),
    project_id: 108,
    project_title: 'Social Media Post Design',
    category: 'Graphic Design',
    client_name: 'Nitin Bansal',
    freelancer_name: 'Kavya Singh',
    budget: 8000,
    status: 'Completed',
    rating: 4.3
  },
  {
    _id: ObjectId('6a8861feab527df55efa3638'),
    project_id: 110,
    project_title: 'AI Chatbot Development',
    category: 'Artificial Intelligence',
    client_name: 'Sahil Khanna',
    freelancer_name: 'Dev Patel',
    budget: 60000,
    status: 'In Progress',
    rating: 4.9
  },
  {
    _id: ObjectId('6a8861feab527df55efa3639'),
    project_id: 109,
    project_title: 'SQL Database Optimization',
    category: 'Database Management',
    client_name: 'Megha Jain',
    freelancer_name: 'Akash Gupta',
    budget: 30000,
    status: 'In Progress',
    rating: 4.7
  },
  {
    _id: ObjectId('6a8861feab527df55efa363a'),
    project_id: 106,
    project_title: 'Machine Learning Model Development',
    category: 'Machine Learning',
    client_name: 'Aditya Mehra',
    freelancer_name: 'Rohit Kumar',
    budget: 55000,
    status: 'In Progress',
    rating: 4.9
  }
]
```

4. Export the documents from the collection into a JSON or CSV file using the mongoexport utility.

```
[sahil@sahilmediasahil]$ touch ~/Desktop/MongoDB/projects2.json
[sahil@sahilmediasahil]$ mongoexport --uri mongodb://localhost:27020 --collection projects --file ~/Desktop/MongoDB/projects2.json --jsonArray=false
total 0
-rw-r--r-- 1 sahilmediasahil staff 0 15 Aug 30:25 projects2.json
[sahil@sahilmediasahil]$ mongoexport --uri mongodb://localhost:27020 --collection projects --file ~/Desktop/MongoDB/projects2.json --jsonArray=false
302w-08-30T18:16:36.173+0530 error writing command line options: bool flag --jsonArray cannot have an argument
302w-08-30T18:16:36.173+0530 try 'mongoexport --help' for more information
[sahil@sahilmediasahil]$ mongoexport --uri mongodb://localhost:27020 --collection projects --file ~/Desktop/MongoDB/projects2.json --jsonArray=false
{"project_id":106,"project_title":"Machine Learning Model Development","category":"Machine Learning","client_name":"Aditya Mehra","freelancer_name":"Rohit Kumar","budget":55000,"status":"In Progress","rating":4.9}
{"project_id":107,"project_title":"WordPress Business Website","category":"Web Development","client_name":"Pooja Arora","freelancer_name":"Vivek Sharma","budget":22000,"status":"Completed","rating":4.5}
{"project_id":108,"project_title":"Social Media Post Design","category":"Graphic Design","client_name":"Nitin Bansal","freelancer_name":"Kavya Singh","budget":8000,"status":"Completed","rating":4.3}
{"project_id":109,"project_title":"SQL Database Optimization","category":"Database Management","client_name":"Megha Jain","freelancer_name":"Akash Gupta","budget":30000,"status":"In Progress","rating":4.7}
{"project_id":110,"project_title":"AI Chatbot Development","category":"Artificial Intelligence","client_name":"Sahil Khanna","freelancer_name":"Dev Patel","budget":60000,"status":"In Progress","rating":4.9}
[sahil@sahilmediasahil]$ mongoexport --uri mongodb://localhost:27020 --collection projects --file ~/Desktop/MongoDB/projects2.csv
302w-08-30T18:16:36.173+0530 connected to: mongodb://localhost/
302w-08-30T18:16:36.173+0530 5 document(s) imported successfully, 0 document(s) failed to import.
[sahil@sahilmediasahil]$ mongoexport --uri mongodb://localhost:27020 --collection projects --file ~/Desktop/MongoDB/projects2.csv
Current MongoDB log is: /var/log/mongodb/mongod.log
Connecting to:
Using MongoDB:
Using mongosh:
mongosh 2.18.0 is available, run 'brew upgrade mongosh' to update
```

5. Create a complete backup of the MongoDB database using the mongodump utility.

```
(base) oshailmadaan@OshailsMacBook ~ % mongodump --db freelanceDB8 --out ~/Desktop/MongoDB/freelance_backup
2026-08-15T18:39:14.647+0538   writing 'freelanceDB8.projects' to '/Users/oshailmadaan/Desktop/MongoDB/freelance_backup/freelanceDB8/projects.bson'
2026-08-15T18:39:15.648+0538   done dumping 'freelanceDB8.projects' (10 documents)
(base) oshailmadaan@OshailsMacBook ~ % ls ~/Desktop/MongoDB/freelance_backup
freelanceDB8
(base) oshailmadaan@OshailsMacBook ~ % ls ~/Desktop/MongoDB/freelance_backup/freelanceDB8
projects.bson      projects.metadata.json
```

6. Delete selected documents or drop the collection/database to demonstrate the need for data restoration.

```
[test> use freelanceDB8
switched to db freelanceDB8
freelanceDB8> show collections
projects
freelanceDB8> db.projects.drop()
true
freelanceDB8> show collections

freelanceDB8> use freelanceDB8
| db.projects.drop()
| show collections
already on db freelanceDB8
freelanceDB8> █
```

7. Restore the backed up database using the mongorestore utility.

```
test> use freelanceDB8
switched to db freelanceDB8
freelanceDB8> show collections
projects
freelanceDB8> db.projects.countDocuments()
10
freelanceDB8> db.projects.findOne()
{
  _id: ObjectId('6a805b7cd04b74cccf6831dd'),
  project_id: 'P104',
  project_title: 'Content Writing',
  client_name: 'EduWorld',
  category: 'Content Writing',
  budget: 12000,
  status: 'Open',
  freelancer_name: 'Priya Singh'
}
```

8. Verify that the database and collection have been restored successfully by displaying: a. The list of databases.

```
[test> show dbs
FreelanceMarketplace 184.00 KiB
admin                40.00 KiB
config              108.00 KiB
freelanceDB8         56.00 KiB
local                80.00 KiB
tests █
```

b. The list of collections in the restored database.

```
[freelanceDB8> show collections  
projects
```

c. The total number of documents in the restored collection.

```
[freelanceDB8> db.projects.countDocuments()  
10
```

d. A sample document from the restored collection.

```
[freelanceDB8> db.projects.findOne()  
{  
  _id: ObjectId('6a805b7cd04b74cccf6831dd'),  
  project_id: 'P104',  
  project_title: 'Content Writing',  
  client_name: 'EduWorld',  
  category: 'Content Writing',  
  budget: 12000,  
  status: 'Open',  
  freelancer_name: 'Priya Singh'  
}
```

9. Create an index on a relevant field in the collection based on the selected business domain. Execute a query that can benefit from the created index and verify the query execution using an appropriate MongoDB command.

```
[freelanceDB8> db.projects.createIndex({ category: 1 })  
category_1  
[freelanceDB8> db.projects.find({ category: "Content Writing" })  
[  
  {  
    _id: ObjectId('6a805b7cd04b74cccf6831dd'),  
    project_id: 'P104',  
    project_title: 'Content Writing',  
    client_name: 'EduWorld',  
    category: 'Content Writing',  
    budget: 12000,  
    status: 'Open',  
    freelancer_name: 'Priya Singh'  
  }  
]
```



```

[freelanceDB8> db.projects.find({
|   category: "Content Writing"
| }).hint({
|   category: 1
| }).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'freelanceDB8.projects',
    parsedQuery: { category: { '$eq': 'Content Writing' } },
    indexFilterSet: false,
    queryHash: 'A248FCC2',
    planCacheShapeHash: 'A248FCC2',
    planCacheKey: '59CAF3CB',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      nss: 'freelanceDB8.projects',
      inputStage: {
        stage: 'IXSCAN',
        nss: 'freelanceDB8.projects',
        keyPattern: { category: 1 },
        indexName: 'category_1',
        isMultiKey: false,
        multiKeyPaths: { category: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: { category: [ ["Content Writing", "Content Writing"] ] }
      }
    }
  },

```

```

    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 1,
    executionTimeMillis: 2,
    totalKeysExamined: 1,
    totalDocsExamined: 1,
    executionStages: {
      isCached: false,
      stage: 'FETCH',
      nReturned: 1,
      executionTimeMillisEstimate: 0,
      works: 2,
      advanced: 1,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      nss: 'freelanceDB8.projects',
      docsExamined: 1,
      alreadyHasObj: 0,
      inputStage: {
        stage: 'IXSCAN',
        nReturned: 1,
        executionTimeMillisEstimate: 0,
        works: 2,
        advanced: 1,
        needTime: 0,
        needYield: 0,
        saveState: 0,
        restoreState: 0,
        isEOF: 1,
        nss: 'freelanceDB8.projects',
        keyPattern: { category: 1 },
        indexName: 'category_1',
        isMultiKey: false,
        multiKeyPaths: { category: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: { category: [ ["Content Writing", "Content Writing"] ] },
        keysExamined: 1,
        seeks: 1,
        dupsTested: 0,
        dupsDropped: 0,
        peakTrackedMemBytes: 0
      }
    }
  }
},

```

